

COMPETITIVE PROGRAMMING

Introduction: Why CP?

It is like a sport.

- Increases your thinking ability.
 - Networking
 - Discipline
 - It's Fun
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Prerequisites:

1. Programming Language (Recommended : C++)

- Basic I/O Problems
- Functions, Loops, Conditional Statements, Patterns, Recursion
- Implementation of Simple English to Code
- Macros, Integers, Overflow, Basic Arithmetic Operations
- Implementing Basic Math and Logic
- Time Complexity

Resources:

- ☐ [Chirag Jain](#) , [Time Complexity](#) (also see docs in description)
- ☐ [Luv Till 16 Videos](#)

Practice:

- ☐ [HackerRank](#) (Till 3 Stars Except STL)

2. STL (In built Data Structures of your preferred Language)

- Vectors,Pairs,Maps,Sets,Strings
- Priority Queue, Dequeue
- Stacks
- Multisets,Unordered Sets
- Ordered Sets
- Pbds

Resources:

- ☐ [Luv STL](#)
- ☐ Chirag Jain 3,4,5,6

Practice:

- ☐ [HackerRank](#)(5 Stars)
- ☐ [HackerEarth CodeMonk Contests](#) 1,2,3,4

3. Basic Math

- GCD,LCM
- Primes,Factors
- Basic Permutation and Combination
- Modular Arithmetic
- Solving Basic Equations

Resources:

- ☐ [CP Algorithms](#)

[After all PreRequisites Solve Problem Solving Of HackerRank Till 5 Stars](#)

[Link](#)

If You Want to go Further (After all the above steps)

- ☐ Start Codeforces By solving 800 Rated Problems
- ☐ See Level 1 of CodeISM 2023
- ☐ Learn Number Theory From CodeNcode
- ☐ Practice Numerous Implementation Problems in order to get Proficient in Programming Language
- ☐ Start Doing Problems

What to Learn Till 1600 Rating ? (What's Rating ? Find out)

- 1.Binary Search
- 2.Prefix,Suffix Sums
3. 2-Pointers,Basic Searching Sorting
- 4.Sliding Window
- 5.Simple Hashing Problems
- 6.Number Theory and Math
7. Recursion
- 8.Basic Dfs,bfs,shortest Paths
- 9.Basic DP
- 10.Basic Combinatorics
- 11.Bit Manipulation

Resources:(Till 1600)

- 1.CodeISM Docs
- 2.[CP Algorithms](#)
- 3.[USACO Guide\(Till Gold or Platinum\)](#)
- 4.[Github Repo](#)
- 5.[Awesome Competitive Programming](#)
- 6.[ShahjalalShohag Blog](#)
- 7.[Mostafa Saad Sheet](#)
- 8.[Bit Hacks](#)
- 9.[Blog Compile on Codeforces](#)
- 10.[Striver Guide](#)

Practice:

1. ABCs of Atcoder
2. CSES Problemset
3. Codechef
4. Codeforces
5. HackerEarth CodeMonks

How To Practice?

■ **[Tutorial] A Way to Practice Competitive Programming.pdf**